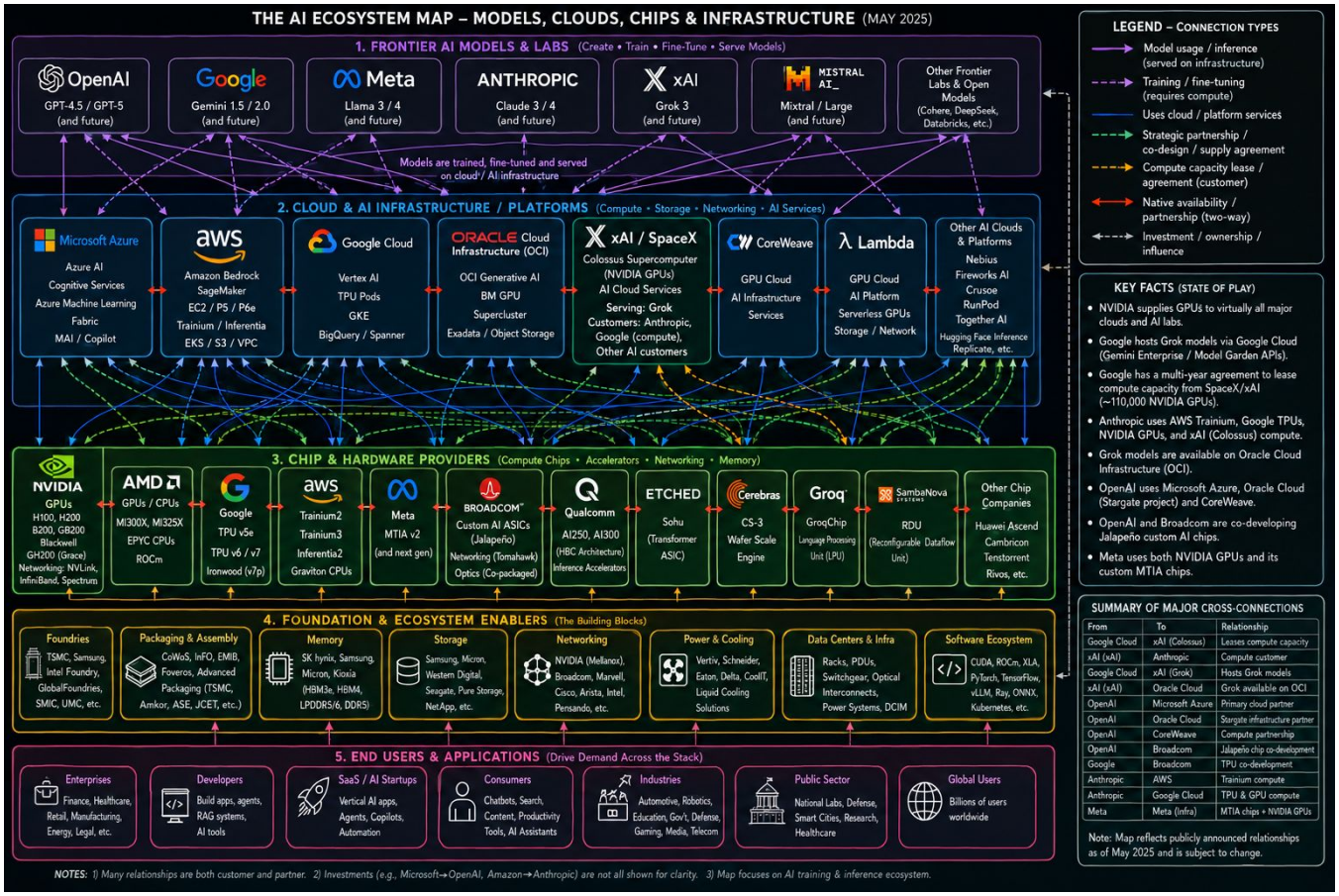
EXPLAINER • OCTOBER 23, 2026

The AI Ecosystem Map, Explained and Updated

A layer-by-layer walk through the “AI Ecosystem Map: Models, Clouds, Chips & Infrastructure”, checked against what is publicly known as of late October 2026, including OpenAI’s Jalapeño chip, Etched’s Frontier Inference Clusters, the SpaceX/xAI compute deals and the 2026 chip roadmaps.



The map being explained. Five horizontal layers (models, clouds, chips, enablers, end users) with a legend, key facts and a cross-connection table on the right.

How to read the status tags

Every dated fact in this guide carries one of four tags so you can tell announcements from rumors:

[OFFICIAL] announced by the company itself (press release, blog, keynote, filing). **[REPORTED]** reputable press citing sources or briefings. **[CLAIM]** a vendor’s own performance or business number, not independently tested. **[UNVERIFIED]** single or secondary source, or sources that conflict.

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The one-paragraph version

AI is a five-layer stack. Labs build models (layer 1) and rent compute from clouds (layer 2). The clouds fill data centers with chips (layer 3), and the chips depend on foundries, memory, packaging, networking, power and software (layer 4). All of it is paid for by demand from businesses, developers and consumers (layer 5). The 2026 story is that the arrows between layers have become tangled: labs now design their own chips (OpenAI Jalapeño), chip startups sell whole data-center racks (Etched), a rocket company rents GPUs to two rival AI labs (SpaceX), and almost every new chip is built for *inference* rather than training. Nvidia still sits under nearly everything, but its share of new purpose-built capacity is now genuinely contested.

ORIENTATION

1. Reading the map

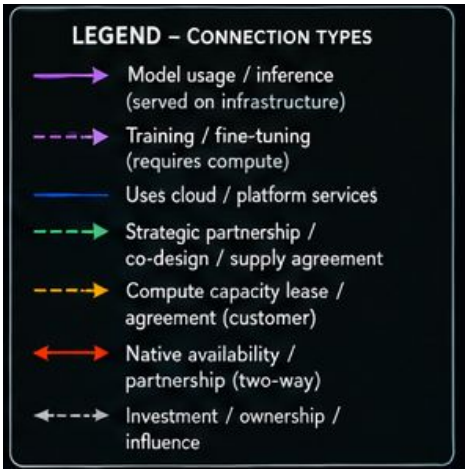
The map is organized as a vertical stack. Each band is a layer of the industry, and arrows show how money, compute and models flow between them. Read it top-down to follow a ChatGPT or Claude request from the model to the silicon it runs on, or bottom-up to follow demand from users through to the factories that make chips.

Layer	What it contains	Role in the stack
1. Frontier models & labs	OpenAI, Google, Meta, Anthropic, xAI, Mistral, others	Create, train, fine-tune and serve models. Biggest buyers of compute.
2. Cloud & AI platforms	Azure, AWS, Google Cloud, Oracle, xAI/SpaceX, CoreWeave, Lambda, neoclouds	Turn chips into rentable compute plus managed AI services.
3. Chips & hardware	Nvidia, AMD, Google TPU, AWS Trainium, Meta MTIA, Broadcom, Qualcomm, Etched, Cerebras, Groq, SambaNova, others	Design the accelerators, CPUs and network silicon.
4. Enablers	Foundries, packaging, memory, storage, networking, power & cooling, data centers, software	The physical and software building blocks every chip depends on.

Layer	What it contains	Role in the stack
5. End users	Enterprises, developers, startups, consumers, industries, public sector, global users	Generate demand that pays for the whole stack.

The legend: seven kinds of relationship

The arrows are color-coded, and the color matters because the same two companies are often linked in several ways at once (Google is a supplier, customer, partner and investor of Anthropic simultaneously).



The map's legend.

Arrow	Meaning	Example from 2026
Solid purple	Model usage / inference served on infrastructure	Claude served to Pro and Max subscribers from SpaceX's Colossus 1
Dashed purple	Training / fine-tuning (needs compute)	GPT-6 Astra pretraining at the Texas Stargate site [REPORTED]
Solid blue	Uses cloud or platform services	OpenAI renting capacity from Azure, Oracle, AWS and Google [REPORTED]
Dashed green	Strategic partnership / co-design / supply	OpenAI and Broadcom co-designing Jalapeño [OFFICIAL]
Dashed orange	Compute capacity lease (customer)	Google leasing about 110,000 Nvidia GPUs from SpaceX [REPORTED]
Red double-headed	Native availability / two-way partnership	Grok models available natively on Vertex AI and OCI [OFFICIAL]
Dashed grey	Investment / ownership / influence	Nvidia's roughly $30B equity stake in OpenAI [REPORTED]

A dating problem worth knowing about

The header says **May 2025**, and a footnote says relationships are "as of May 2025". That cannot be fully right. Several items on the map only became public a year later:

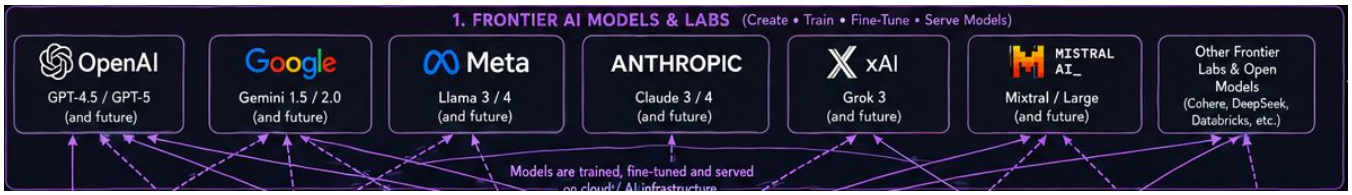
- **xAI / SpaceX as one entity.** SpaceX absorbed xAI in an all-stock deal that closed in early February 2026 [REPORTED].
- **Anthropic as a Colossus customer.** Anthropic agreed to rent all of Colossus 1 on May 6, 2026 [REPORTED].
- **Google leasing about 110,000 GPUs from SpaceX/xAI.** Disclosed in SpaceX's SEC filing in June 2026 [REPORTED].
- **The name "Jalapeño".** OpenAI and Broadcom revealed it on June 24, 2026 [OFFICIAL].

- **Qualcomm AI250 / AI300 with "HBC architecture".** HBC is Qualcomm's 2026 near-memory design; AI300 is a 2028 part [OFFICIAL].

Meanwhile, the model names on the top row (GPT-4.5, Gemini 1.5/2.0, Claude 3/4, Llama 3/4, Grok 3) are spring-2025 vintage. The most likely explanation is a 2025 graphic partly refreshed in mid-2026 without updating the header or the model row. Treat the *relationships* on the map as roughly mid-2026 and the *model versions* as stale. The rest of this guide brings both up to October 2026.

LAYER 1

2. Layer 1: Frontier AI models and labs



Layer 1 as drawn on the map.

This layer is where models are created. Labs need enormous compute to *train* a model once, and then even more, in aggregate, to *serve* it to hundreds of millions of users. That second cost, inference, is what drives most of the 2026 hardware news in later sections. A useful mental model: every lab on this row is now also a major compute buyer, and several are becoming chip designers.

OpenAI

ON THE MAP

GPT-4.5 / GPT-5 (and future).

WHERE IT STANDS, OCT 2026

- **GPT-5.5** ("Spud") launched Apr 23, 2026; GPT-5.5 Instant became the free default May 5; a gated GPT-5.5-Cyber preview followed May 7 [REPORTED].
- **GPT-6 family:** Astra (limited preview Sept 3, paid users Sept 4), Sol and Luna (Sept 22), and GPT-6 Instant with "Intelligent UI" (Oct 7 paid, Oct 8 free) [REPORTED].
- GPT-6 Astra is reported to be OpenAI's largest pretraining run, its first on more than 100k GPUs, at the Texas Stargate site [REPORTED].
- Compute is now spread across Azure, Oracle/Stargate, CoreWeave, AWS, Google Cloud, Cerebras, AMD and its own Broadcom-built Jalapeño chip (section 5).

Google / DeepMind

ON THE MAP

Gemini 1.5 / 2.0 (and future).

WHERE IT STANDS, OCT 2026

- **Gemini 3.1 Pro** on Feb 19, 2026; **Gemini 3.8 Flash** (Sept 2) and **3.8 Live** (Sept 24) [REPORTED].
- **Gemini 4 Argon:** early access for cyber defenders Sept 30, then general availability and Gemini app rollout on Oct 13, 2026 [REPORTED].
- Google is the only lab that owns every layer: models, a cloud, its own chips (TPU) and its own data centers. It still buys Nvidia GPUs, and in 2026 even rented GPUs from SpaceX (section 3).

Meta

ON THE MAP

Llama 3 / 4 (and future).

WHERE IT STANDS, OCT 2026

- Meta Superintelligence Labs launched **Muse Spark** on Apr 8, 2026, with **closed weights**, replacing Llama as Meta's main line [OFFICIAL].
- Follow-ups: Muse Image (July 7) and Muse Spark 1.1 (July 9) [REPORTED]; Muse Wonder on Oct 12 [UNVERIFIED]. No Llama 5 has been released.
- This is a big strategic shift: Meta was the flagship of open-weight models in 2024-2025. Open weights now come mostly from Chinese labs (DeepSeek, Qwen).

Anthropic

ON THE MAP

Claude 3 / 4 (and future).

WHERE IT STANDS, OCT 2026

- 2026 releases: Opus 4.6 (Feb 5), Sonnet 4.6 (Feb 17), Opus 4.7 (Apr 16), Opus 4.8 (May 28), **Opus 5** (July 24) [REPORTED].
- Gated **Claude Mythos** cyber program from Apr 7; Fable 5.1 and Mythos 5.1 around Sept 1 [REPORTED].
- **Claude Opus 5.5** (Sept 22, 2026): roughly Fable 5.1-level on most work, about 40% cheaper to run than Opus 5, output over 30% faster, priced at $4 / $20 per million input / output tokens [OFFICIAL]. Sonnet 5.5 (Sept 28) and Haiku 5.5 (Oct 7) followed [REPORTED].
- Anthropic is the most multi-sourced lab on the map: AWS Trainium, Google TPUs, Nvidia GPUs, SpaceX's Colossus and, from 2027, AMD MI450s.

xAI

ON THE MAP

Grok 3 (and future).

WHERE IT STANDS, OCT 2026

- xAI is no longer a standalone company: SpaceX absorbed it in February 2026 at a reported combined value of about $1.25 trillion [REPORTED].
- Grok 4.1 Fast and Grok 4.20 are listed on Google's Vertex AI [OFFICIAL]; Grok 4.7 reportedly shipped Sept 21, 2026 [UNVERIFIED].
- **Grok 5 has not shipped.** It slipped from late 2025 to Q1 then Q2 2026, and was last officially described as "in training" [REPORTED].
- xAI moved its main training to Colossus 2, which freed Colossus 1 to be rented to Anthropic and Google (section 3) [REPORTED].

Mistral AI

ON THE MAP

Mixtral / Large (and future).

WHERE IT STANDS, OCT 2026

- Mistral's 2026 story is infrastructure: $830M of debt (its first) for a 44 MW site at Bruyères-le-Châtel near Paris with 13,800 Nvidia GB300 GPUs, due online Q2 2026 [REPORTED].
- Targets: 200 MW across Europe by end-2027 and 1 GW by 2030, serving itself and third parties; a second site in Sweden (about €1.2B) [REPORTED].
- An anchor group of European companies (ASML, CMA-CGM, Caisse des Dépôts, Capgemini, Amadeus) was named in Aug 2026 [REPORTED]. This research did not verify a specific 2026 Mistral model release.

Other labs & open models

ON THE MAP

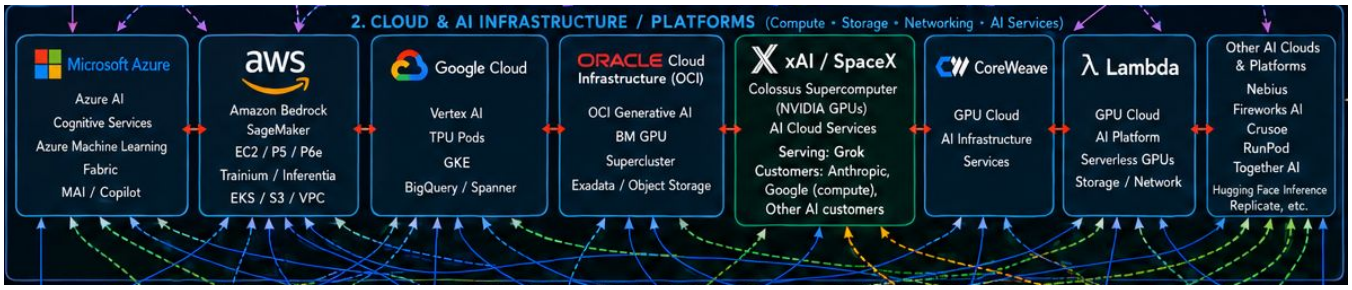
Cohere, DeepSeek, Databricks, etc.

WHERE IT STANDS, OCT 2026

- **DeepSeek V4** preview, Apr 24, 2026: V4-Pro and V4-Flash (284B), MIT license, 1M-token context, with support for Huawei's Ascend 950 chips [REPORTED].
- **Qwen 3.8-Max** (Alibaba), Aug 3, 2026: a 2.4-trillion-parameter mixture-of-experts model with 95B active parameters; weights released Aug 12 under a custom license [REPORTED].
- Chinese labs are now the main source of frontier-class open weights, and DeepSeek's Ascend support is a sign of a separate Chinese hardware stack forming (section 10). Cohere and Databricks were not re-researched for this guide.

LAYER 2

3. Layer 2: Cloud and AI infrastructure platforms



Layer 2 as drawn on the map.

Clouds turn chips into something you can rent by the hour, the token or the gigawatt-year. There are three kinds on the map: **hyperscalers** (Azure, AWS, Google Cloud, Oracle), which also build their own chips and model platforms; **neoclouds** (CoreWeave, Lambda, Nebius, Crusoe and others), which are mostly Nvidia GPU landlords; and one oddity, **xAI / SpaceX**, a lab whose spare supercomputer became a cloud business.

Microsoft Azure

ON THE MAP

Azure AI, Azure ML, Fabric, MAI / Copilot.

WHERE IT STANDS, OCT 2026

- **First cloud to power up Nvidia Vera Rubin NVL72**, announced at GTC on Mar 16, 2026 [OFFICIAL].
- **Maia 200**, Microsoft's own inference chip, announced Jan 26, 2026 and deployed in US Central (Iowa) and then US West 3 (Arizona) [OFFICIAL]. OpenAI's GPT-5.2 is reported to run on it [REPORTED]. Maia is missing from the map's chip layer.
- Named by OpenAI as a data-center partner for gigawatt-scale Jalapeño deployment [OFFICIAL].
- The map calls Azure OpenAI's "primary cloud partner". That was true in 2024; in 2026 Azure is one of several (see section 9).

AWS

ON THE MAP

Bedrock, SageMaker, EC2 P5/P6e, Trainium/Inferentia, EKS/S3/VPC.

WHERE IT STANDS, OCT 2026

- **Trainium3** UltraServers generally available Dec 2, 2025: 144 chips per system, 362 FP8 PFLOPS, 4.4x the compute of Trainium2 [OFFICIAL]. **Trainium4** is in development, with NVLink Fusion support [OFFICIAL].
- **Anthropic deal (Apr 2026):** up to 5 GW across Trainium 2/3/4 and more than $100B of AWS spend over 10 years, with nearly 1 GW online by end-2026 [OFFICIAL]. Project Rainier is about 500k Trainium2 chips [REPORTED].
- At GTC 2026 AWS said it would deploy more than 1M Nvidia GPUs including Rubin, plus "Nvidia Groq 3 LPUs" [OFFICIAL].
- A $50B investment by Amazon in OpenAI plus Trainium capacity has been reported [REPORTED]; this research did not find the primary announcement.

Google Cloud

ON THE MAP

Vertex AI, TPU Pods, GKE, BigQuery/Spanner.

WHERE IT STANDS, OCT 2026

- **TPU v7 Ironwood** GA Nov 6, 2025; **TPU 8t** (training) and **TPU 8i** (inference) announced Apr 22, 2026, GA later in 2026 [OFFICIAL]. Details in section 4.
- **Hosts Grok**: xAI models are offered as managed partner APIs on Vertex AI via Model Garden [OFFICIAL].
- **Rents GPUs from SpaceX:** about 110,000 Nvidia GPUs at Memphis and Southaven for $920M per month, Oct 2026 to June 2029 (about $30.3B if fully delivered), with exit clauses if SpaceX misses delivery by Sept 30, 2026 [REPORTED, from SpaceX's SEC filing]. Google called it "bridge capacity" for Gemini Enterprise demand.
- Anthropic has access to up to 1M TPUs, with well over 1 GW coming online in 2026 [OFFICIAL].

Oracle Cloud (OCI)

ON THE MAP

OCI Generative AI, bare-metal GPU, Supercluster, Exadata.

WHERE IT STANDS, OCT 2026

- OpenAI's main Stargate build partner. Abilene, Texas targets 1.2 GW with full build in Q4 2026; plans to expand it to 2.1 GW were dropped **[REPORTED, Epoch AI]**.
- Grok has been available in OCI Generative AI since June 17, 2025 **[OFFICIAL]**.
- Nvidia Rubin was deployed at OCI by Aug-Sept 2026 **[REPORTED]**.

xAI / SpaceX

ON THE MAP

Colossus supercomputer (Nvidia GPUs). Serving Grok; customers Anthropic, Google.

WHERE IT STANDS, OCT 2026

- Colossus 1 in Memphis is the supercomputer xAI built in 2024. After xAI moved training to Colossus 2, its capacity was rented out **[REPORTED]**.
- **Anthropic** (May 6, 2026): all of Colossus 1, more than 300 MW and more than 220,000 Nvidia GPUs, aimed at Claude Pro and Max subscribers. Anthropic immediately doubled five-hour Claude Code limits **[REPORTED]**. A $1.25B-per-month price has been reported but not corroborated **[UNVERIFIED]**.
- **Google** (June 2026): about 110,000 GPUs, $920M per month (see Google Cloud above) **[REPORTED]**.
- Why it matters: compute has become a tradable commodity, to the point that rival labs rent from each other. SpaceX has also floated orbital data centers, and Anthropic expressed interest **[REPORTED]**.

CoreWeave

ON THE MAP

GPU cloud, AI infrastructure services.

WHERE IT STANDS, OCT 2026

- The largest neocloud. OpenAI contracts total up to about $22.4B (March, May and September 2025 agreements) **[OFFICIAL]**.
- About 590 MW across six Core Scientific sites **[REPORTED]**. Among the first clouds with Nvidia Rubin in 2026 **[REPORTED]**.

Lambda & other neoclouds

ON THE MAP

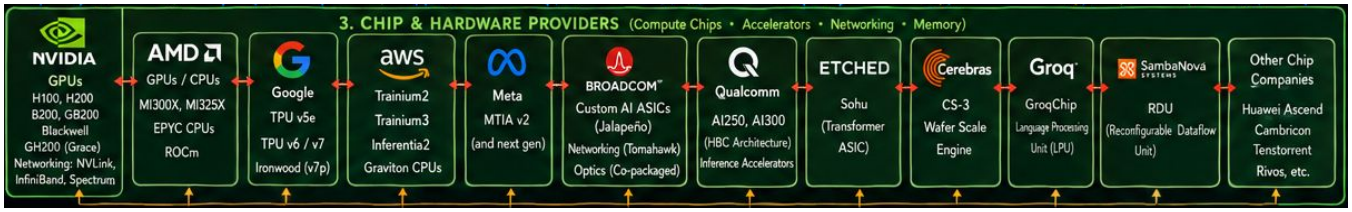
Lambda; Nebius, Fireworks AI, Crusoe, RunPod, Together AI, Hugging Face, Replicate.

WHERE IT STANDS, OCT 2026

- Two different businesses are mixed in this box. **Infrastructure neoclouds** (Lambda, Nebius, Crusoe, RunPod) rent GPUs. **Inference platforms** (Fireworks, Together, Hugging Face, Replicate) sell model serving per token, often on top of the former.
- Nebius was among the first to deploy Nvidia Rubin **[REPORTED]**. This guide did not re-verify 2026 financing or IPO news for the rest.

LAYER 3

4. Layer 3: Chip and hardware providers



Layer 3 as drawn on the map.

This is the layer that changed most between the map's original 2025 snapshot and today. Three categories now matter:

- **Merchant GPUs** sold to anyone: Nvidia and AMD.
- **Hyperscaler and lab custom silicon (ASICs)**: Google TPU, AWS Trainium, Microsoft Maia, Meta MTIA and OpenAI Jalapeño, mostly designed with **Broadcom** (or MediaTek / Marvell) and made at TSMC.
- **Inference specialists**: Etched, Cerebras, Groq, SambaNova, Qualcomm, Tenstorrent, plus China's Huawei and Cambricon.

The big shift: in 2024 the question was “who can get Nvidia GPUs?”. In 2026 it is “who can serve tokens most cheaply per watt?”. Training still runs overwhelmingly on Nvidia (plus TPUs and Trainium), but nearly every new chip launched in 2026 is aimed at inference.

Nvidia: still the center of gravity

The map lists H100, H200, B200, GB200, Blackwell and GH200. Those are now the installed base. The current roadmap is:

Generation	Timing	Key facts
Blackwell Ultra (GB300 NVL72)	Shipping since H2 2025	The baseline OpenAI benchmarked Jalapeño against. DGX Station GB300: 748 GB, 20 PFLOPS FP4 [OFFICIAL].
Vera Rubin (NVL72)	Full production May/June 2026	Presented at GTC 2026 as 7 chips and 5 rack-scale systems, including the Vera CPU and BlueField-4 [OFFICIAL]. TSMC 3nm class, HBM4, about 50 PFLOPS FP4 per GPU vs 20 for Blackwell [REPORTED]. Live at Azure (first), CoreWeave, Google Cloud, OCI and Nebius by Aug-Sept 2026 [REPORTED].
Rubin Ultra (Kyber rack)	2027	About 100 PFLOPS FP4 [OFFICIAL roadmap]. Nvidia is reportedly testing pared-down 192 GB configurations because of HBM shortages [REPORTED].
Feynman	About 2028	New Rosa CPU, LP40 LPU (from the Groq-derived team), BlueField-5, CX10, Kyber with co-packaged optics [OFFICIAL as platform]. TSMC A16 and 3D stacking reported, not confirmed [UNVERIFIED].

The Groq deal. On Dec 24, 2025 Nvidia signed a non-exclusive license to Groq’s inference technology and hired its founder Jonathan Ross and president Sunny Madra [OFFICIAL]. Groq stays independent and GroqCloud keeps running. CNBC put the value at about $20B citing an investor [REPORTED]. This is why “Groq 3 LPUs” now appear inside Nvidia announcements and why an LPU sits in the Feynman platform. It is Nvidia buying an inference-specialist architecture rather than competing with it.

The OpenAI investment. The Sept 2025 letter of intent for up to $100B tied to 10 GW was replaced in Feb 2026 by a direct equity stake of about $30B in OpenAI’s $110B round [REPORTED]. Jensen Huang later said the $100B was “probably not in the cards” [REPORTED].

AMD: the second merchant source

The map shows MI300X/MI325X. AMD’s 2026 story is the **MI450** GPU and the **Helios** rack, and a run of gigawatt-scale deals that the map does not draw at all:

Customer	Announced	Deal
OpenAI	Oct 6, 2025	6 GW across generations; first 1 GW of MI450 in H2 2026. AMD granted OpenAI a warrant for up to 160M shares (about 10%) that vests with deployment [OFFICIAL]. Whether the first GW is live was not confirmed [UNVERIFIED].
Meta	2026	6 GW; first GW in H2 2026 [OFFICIAL].
Anthropic	July 22, 2026	Up to 2 GW of MI450 / Helios; first GW in H1 2027 [OFFICIAL].

Hyperscaler custom silicon

Chip	Status (Oct 2026)	Notable specs and facts
Google TPU v7 Ironwood	GA Nov 6, 2025	9,216-chip pods linked by optical circuit switching [OFFICIAL]. The map’s “Ironwood (v7p)”.

Chip	Status (Oct 2026)	Notable specs and facts
Google TPU 8t / 8i	Announced Apr 22, 2026; GA later 2026	First split into separate training (8t) and inference (8i) chips. 8t: 9,600-chip superpod, about 121 FP4 exaflops, 2 PB shared memory, up to 2.8x training price-performance vs Ironwood. 8i: 80% better inference performance per dollar [OFFICIAL]. Broadcom (8t) and MediaTek (8i) as design partners [UNVERIFIED].
AWS Trainium3	GA Dec 2, 2025	3nm; 144 chips per UltraServer, 362 FP8 PFLOPS, 20.7 TB HBM3e [OFFICIAL].
AWS Trainium4	In development	Targets 6x FP4, 3x FP8 and 4x memory bandwidth vs Trainium3; NVLink Fusion [OFFICIAL]. Late 2026 / early 2027 date [UNVERIFIED].
Microsoft Maia 200	Deployed (Iowa, Arizona)	TSMC N3, about 10 PFLOPS FP4, 216 GB HBM3e at 7 TB/s, 272 MB SRAM, 750 W, clusters of up to 6,144 chips [OFFICIAL]. Microsoft claims 3x the FP4 of Trainium3 [CLAIM]. **Not on the map.**
Meta MTIA	Roadmap through 2027	MTIA 300/400/450/500 on a roughly 6-month chiplet cadence [OFFICIAL]. Broadcom partnership extended through 2029, more than 1 GW in its first phase [OFFICIAL].

Notice the pattern: Broadcom is behind most of these. Broadcom does not sell an “AI chip” of its own; it turns a customer’s architecture into manufacturable silicon, supplies the networking (Tomahawk Ethernet switches) and handles the TSMC relationship. The map’s Broadcom tile (“Custom AI ASICs (Jalapeño), Networking (Tomahawk), Optics (Co-packaged)”) captures this well.

Inference challengers

Company	On the map	Where it stands, Oct 2026
Etched	Sohu (transformer ASIC)	Now sells rack-scale “Frontier Inference Clusters” and has dropped the transformer-only pitch. Valuation went from $10.3B (July) to about $21B (Aug), with offers above $40B reported in Oct. See section 6.
Cerebras	CS-3 wafer-scale engine	Public company: Nasdaq IPO (CBRS) on May 14, 2026 [OFFICIAL]. OpenAI deal (Jan 2026) for 750 MW with an option for 1.25 GW more through 2030; OpenAI holds warrants and lent Cerebras $1B [OFFICIAL]. G42/MBZUAI were 86% of 2025 revenue [OFFICIAL, S-1].
Groq	GroqChip LPU	Technology licensed to Nvidia and leadership hired (Dec 2025); Groq continues independently under Simon Edwards with GroqCloud [OFFICIAL].
SambaNova	RDU (reconfigurable dataflow)	SN50 chip and $350M+ Series E with an Intel partnership (Feb 24, 2026) after Intel acquisition talks stalled [REPORTED]. $1B Series F first close at $11B (July 8, 2026) [REPORTED]. SN50 shipping H2 2026, SoftBank first [REPORTED].
Qualcomm	AI250, AI300 (HBC architecture)	HBC (High Bandwidth Compute) puts a compute die directly under stacked LPDDR memory instead of using HBM. Qualcomm claims 133 TB/s effective bandwidth per AI250 card and up to 6x better bandwidth per watt than HBM [CLAIM]. Peak FLOPS not disclosed. AI250 sampling mid-2027 or later; AI300 (HBC Gen 2) about 2028 [REPORTED].
Tenstorrent	Under “Other”	Galaxy Blackhole server GA Apr 28, 2026 (32 chips, 23 PFLOPS FP8); $1.5B Series E at $6.2B post-money on Oct 19, 2026 [OFFICIAL].
Huawei Ascend	Under “Other”	Ascend 950PR reportedly in mass production from Mar 2026 [UNVERIFIED]. Atlas 950 SuperPoD (8,192 chips) planned for Q4 2026 and Atlas 960 (15,488) for Q4 2027 [REPORTED]. Huawei compensates for weaker chips with scale.

FEATURED CHIP

5. Deep dive: OpenAI Jalapeño

Jalapeño is OpenAI's first in-house chip, which it calls an "Intelligence Processor". It is an **inference-only** ASIC for running ChatGPT, Codex, the API and agent products. It does not train models; training stays on Nvidia and AMD. On the map it appears inside Broadcom's tile and in the "OpenAI → Broadcom: Jalapeño chip co-development" row.

How it came about

Date	Event
Oct 2024	Reuters reports OpenAI is designing its first chip with Broadcom and TSMC, and scaling back plans for its own foundries [REPORTED].
Early 2025	A team of about 40 led by Richard Ho (formerly of Google's TPU team) is finalizing a TSMC 3nm systolic-array design with HBM [REPORTED].
Oct 13, 2025	OpenAI and Broadcom announce they will jointly deploy **10 GW** of custom accelerators, rolling out through 2029 [OFFICIAL]. An analyst estimate of about $350B for the deal is not confirmed [UNVERIFIED].
June 24, 2026	**Public reveal** of Jalapeño. Engineering samples are already running workloads, including GPT-5.3-Codex-Spark, at target clock and power. Broadcom stock rises about 3.4% [OFFICIAL].
Aug 25, 2026	Detailed architecture shown at **Hot Chips 2026** [OFFICIAL presentation].
End of 2026	Initial deployment in OpenAI's infrastructure; gigawatt-scale deployment with partners such as Microsoft to follow [OFFICIAL]. Volume in 2027 [REPORTED].

Who builds what

- **OpenAI:** architecture and the software stack, including a programming model called "Gluon" [REPORTED].
- **Broadcom:** turns the design into silicon and supplies the networking, including Tomahawk Ethernet switches [OFFICIAL].
- **Celestica:** boards, racks and system integration [OFFICIAL].
- **TSMC:** manufacturing on 3nm [REPORTED]. **Samsung** is reported as the HBM4 supplier, from Korean media only [UNVERIFIED].

Specifications disclosed at Hot Chips 2026

All numbers are OpenAI's own and have not been independently tested.

Spec	Jalapeño (per package)	Context
Process	TSMC 3nm, first generation	Same node class as Rubin, Maia 200 and Trainium3
Memory	216 GB HBM4 at 15.4 TB/s (six stacks per most reports)	Same capacity as Maia 200 (216 GB) but HBM4, about 2x the bandwidth
Compute	About 13.4 PFLOPS MXFP4; about 3.4 PFLOPS MXFP8	Rubin is about 50 PFLOPS FP4 (sparse); Maia 200 about 10
Power	700 W rated; at or under 550 W sustained on tested workloads	GB200 / GB300 accelerators are about 1,200 / 1,400 W
Clock	1.70 GHz on lab silicon, 1.80 GHz planned	
Architecture	Spatial design: 64 core slices, each paired with a local HBM slice	Large on-chip SRAM suspected, not confirmed
Die size	About 840 mm², near the reticle limit	[UNVERIFIED]: single secondary source

Spec	Jalapeño (per package)	Context
Rack	128 chips; about 1.7 EFLOPS FP4; about 27.5 TB HBM4	Rack bandwidth just under 2 PB/s
Pod	2,048 chips in 16 racks; about 27 EFLOPS; 432 TiB HBM4	About 32 PB/s aggregate memory bandwidth

Performance claims, and how to read them

- [CLAIM] 1.5 to 1.9x more work per watt than Nvidia GB200 / GB300 NVL72 at peak throughput.
- [CLAIM] 1.7 to 3.6x lower end-to-end latency, and 2.1 to 4.1x better on ultra-low-latency workloads.
- [CLAIM] About 50% lower inference cost, per Bloomberg's reporting of OpenAI's figures.
- Models tested: gpt-oss-120B, DeepSeek R1 and Kimi K2.5. Results were normalized to the accelerators' TDP.

Caveats

The comparison is against Blackwell-generation racks, not Nvidia's current Vera Rubin. The Register notes the Nvidia numbers may come from an unofficial InferenceX run, and that speculative decoding was excluded. There are no independent benchmarks, no price, and no transistor count or interconnect bandwidth. Whether outside companies can ever buy Jalapeño is unclear; the companies only said it was "built for current and future LLMs across the industry".

Why it matters

- **AI helped build the chip.** Design to tape-out took about nine months, which is very fast. OpenAI says GPT-6 Astra and Codex wrote and optimized low-level kernels, some running 1.5 to 1.8x faster than human-expert code on attention and mixture-of-experts blocks [CLAIM].
- **Bargaining power.** Even if Jalapeño serves only part of OpenAI's traffic, a credible in-house alternative changes how OpenAI negotiates GPU prices with Nvidia and AMD.
- **A multi-generation roadmap.** Broadcom's CEO Hock Tan called it "just the beginning". Gen 2 is reported to be deep in development and could tape out within months; Gen 3 is underway [REPORTED].
- **Inference is the bill.** Serving hundreds of millions of users costs far more over a model's life than training it once, so a chip that halves serving cost goes straight to OpenAI's margins.

FEATURED COMPANY

6. Deep dive: Etched and the Frontier Inference Clusters

On the map Etched is a single small tile: "Sohu (Transformer ASIC)". That describes the company as of 2024. In 2026 Etched re-emerged with a different product and became one of the most valuable chip startups in the world.

From Sohu to Frontier Inference Clusters

- **June 2024:** Sohu announced as a chip that only runs transformer models, trading flexibility for speed. Its "20x faster than an H100" claim was never independently shown and is now outdated [CLAIM].
- **June 30, 2026:** Etched comes out of stealth with **Frontier Inference Clusters (FIC)** [OFFICIAL]. Instead of selling chips, it sells complete rack-scale systems: silicon, an HBM memory subsystem, custom racks, cooling, networking and software.
- It is now pitched as able to run any frontier model, including mixture-of-experts models like DeepSeek and Qwen. That quietly walks back the original transformer-only framing [REPORTED].

The two named technologies

Technology	What Etched says it does
Low-Voltage Inference (LVI)	The math blocks run at under half the voltage of typical AI chips. Because power rises steeply with voltage, this lets Etched pack “multiple times” more FLOPs into the same power budget. Etched says it sustains more than 80% of peak FLOPs on trillion-parameter sparse MoE models without throttling [CLAIM].
Cluster-Scale Memory (CSM)	A hybrid HBM / SRAM memory pool shared across the whole scale-up domain over a proprietary interconnect, so a very large model can sit in fast memory spread across many chips [CLAIM].

The silicon (A0 revision) taped out on **TSMC N4P** (4nm) and worked first time [OFFICIAL]. That is a generation behind Jalapeño and Rubin (3nm), so Etched is betting on architecture and voltage, not process node. **No public spec sheet exists**: no HBM capacity, TOPS, power or clock figures have been disclosed.

The money

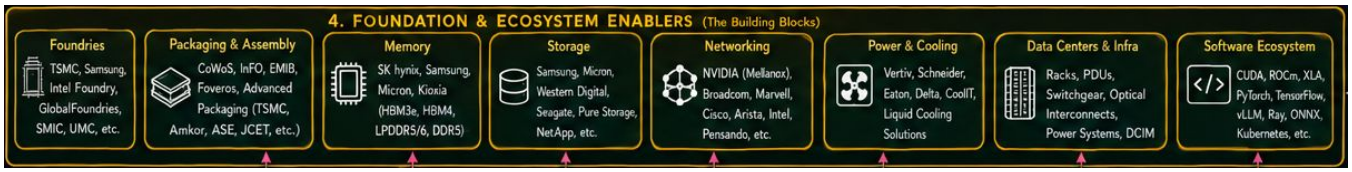
Date	Event
June 30, 2026	Over $1B in signed contracts (customers mostly undisclosed); $800M raised across previously unannounced rounds; 400+ engineers, a factory in Taiwan, and a San Jose lab with its own 2 MW data center [CLAIM].
July 23, 2026	$300M Series C at a **$10.3B** valuation, led by Sequoia with a16z, Jane Street, Diffusion and SK Hynix [OFFICIAL].
Aug 18, 2026	$700M at about **$21B**, led by Jane Street, bringing total funding to about $1.9B. Tied to the first rack shipment, to Jane Street, its first named customer [REPORTED]. The shipment itself is lightly corroborated [UNVERIFIED].
Oct 2026	Reported to be fielding investor offers at more than **$40B** [UNVERIFIED].
2027	Volume production ramp targeted [REPORTED].

How to read Etched

Etched’s valuation roughly quadrupled in three months on the strength of contracts and a first customer, not benchmarks. Jane Street, a trading firm, is an unusual first customer; it reportedly tested a cluster in its own data center before buying. The open questions are the same as for Jalapeño, only larger: no independent performance data, no published specs, and a customer list that is mostly private. The shift to selling whole systems puts Etched in the same business as Nvidia’s NVL72 racks rather than as a component supplier.

LAYER 4

7. Layer 4: Foundation and ecosystem enablers



Layer 4 as drawn on the map.

These are the building blocks that every chip on layer 3 depends on. In 2026 several of them, not the chips themselves, are the real limits on how fast AI capacity can grow.

Foundries: TSMC, Samsung, Intel Foundry, GlobalFoundries, SMIC, UMC

Almost every leading AI chip is made by **TSMC**: Rubin, Jalapeño, Maia 200 and Trainium3 on 3nm-class nodes, Etched on N4P, and Nvidia's Feynman reportedly on A16 [UNVERIFIED]. That concentration in Taiwan is the single biggest supply-chain risk on the map. SambaNova's 2026 partnership with Intel is a small sign of interest in alternatives [REPORTED]. China's SMIC makes Huawei's chips, but US limits on chipmaking equipment constrain how many Ascend 950s it can produce [REPORTED].

Packaging: CoWoS, InFO, EMIB, Foveros

Modern AI chips are not one piece of silicon. Compute dies and HBM stacks sit side by side on a silicon interposer (TSMC's CoWoS is the standard). Packaging capacity has been a bottleneck since 2023, and every new 2026 design (Jalapeño with six HBM4 stacks, Rubin with HBM4, Meta's chiplet-based MTIA) adds demand for it.

Memory: SK hynix, Samsung, Micron (HBM3e, HBM4, LPDDR5/6)

High-bandwidth memory is the binding constraint of 2026. Suppliers report HBM capacity committed through 2027 [REPORTED], and Nvidia is reportedly testing Rubin Ultra configurations with less memory because of it [REPORTED]. Inference makes this worse: generating each token means streaming the model's active weights out of memory, so memory bandwidth, not FLOPs, often limits speed. Three responses are visible:

- **More, faster HBM4:** Jalapeño (216 GB, 15.4 TB/s) and Rubin.
- **Avoiding HBM:** Qualcomm's HBC stacks cheaper LPDDR directly on compute [CLAIM]; Etched mixes HBM and SRAM in a shared pool [CLAIM].
- **Memory makers investing upstream:** SK Hynix is an investor in Etched's Series C [OFFICIAL].

Networking: Nvidia (Mellanox), Broadcom, Marvell, Cisco, Arista

Large models are split across thousands of chips, so the network between them is part of the computer. Two camps exist: Nvidia's NVLink and InfiniBand, and Ethernet led by Broadcom's Tomahawk switches (used in Jalapeño racks). Google uses optical circuit switching in TPU pods, Maia 200 builds the network interface into the chip, and Etched uses a proprietary interconnect. Next is **co-packaged optics**, light-based links built into the chip package, which Nvidia's Feynman platform includes.

Power, cooling and data centers

In 2026 deals are measured in **gigawatts**, not chips. One gigawatt is roughly the output of a large nuclear reactor. OpenAI alone has announced Nvidia 10 GW (since restructured), AMD 6 GW, Broadcom 10 GW and Cerebras 0.75 GW plus options. Building that capacity is much slower than announcing it:

Stargate site	Target	Timing
Abilene, TX (Oracle)	1.2 GW	Full build Q4 2026; about 0.3 GW was operating in April 2026

Stargate site	Target	Timing
Shackelford County, TX	2.0 GW	Q4 2028
Doña Ana County, NM	2.2 GW	Q4 2028
Milam County, TX	1.2 GW	Q4 2028
Port Washington, WI	1.3 GW	Q4 2028
Saline Township, MI	1.4 GW	Q4 2028; construction underway by June 2026
Lordstown, OH	Under 0.3 GW	

Source: Epoch AI's April 2026 site tracker, which found about 0.3 GW operating against about 9.3 GW projected across seven sites. OpenAI said in April 2026 that it had passed its 10 GW US target, but that figure is mostly contracted, not built **[REPORTED]**. In March 2026 OpenAI also restructured Stargate toward renting cloud capacity rather than building everything itself **[REPORTED]**. Efficiency per watt (Jalapeño's 700 W vs about 1,400 W for a GB300 GPU) matters precisely because power, not money, is the scarce input.

Software ecosystem: CUDA, ROCm, XLA, PyTorch, TensorFlow, vLLM, Ray, ONNX, Kubernetes

Software is why Nvidia's lead has lasted: CUDA has two decades of kernels, libraries and developer habit behind it. Every challenger needs its own stack (AMD ROCm, Google XLA/JAX, AWS Neuron, OpenAI's Gluon). Open serving engines like vLLM matter because they let one model run on many chips. The 2026 twist is that models now write some of this software themselves, as with Jalapeño's AI-written kernels, which may erode CUDA's moat faster than human porting could.

LAYER 5

8. Layer 5: End users, applications and the agent layer



Layer 5 as drawn on the map.

The bottom layer is demand: enterprises, developers, startups, consumers, industries, the public sector and billions of global users. The map shows arrows rising from here into every layer above, because this is ultimately what pays for the chips. Three 2026 developments change what this layer looks like:

- **Agents, not chats.** Usage has shifted from single questions to long-running agents that call tools and run for minutes or hours. Agents use far more tokens per task, which is a major reason inference demand keeps outrunning supply.
- **Open plumbing for agents.** On Dec 9, 2025 Anthropic, OpenAI and Block co-founded the **Agentic AI Foundation** under the Linux Foundation. Anthropic donated the Model Context Protocol (MCP); Block's goose and OpenAI's AGENTS.md were founding projects; Google, Microsoft, AWS, Cloudflare and Bloomberg are backers **[OFFICIAL]**. A 2026-07-28 MCP release candidate makes the protocol stateless, with breaking changes **[OFFICIAL]**. Google's A2A protocol covers agent-to-agent communication, while MCP covers agent-to-tool.
- **Gated tiers for dangerous capabilities.** All three leading labs now release cyber-capable models first to vetted defenders: GPT-5.5-Cyber, Claude Mythos and Gemini 4 Argon's cyber-first rollout **[REPORTED]**.

A concrete example of how layer 5 drives layer 2: Anthropic said the Colossus 1 capacity would go directly to Claude Pro and Max subscribers, and on the same day doubled five-hour Claude Code limits and removed peak-hour throttling **[REPORTED]**. Consumer and developer demand directly triggered a 300 MW compute lease.

VERIFICATION

9. Fact-checking the map's Key Facts and connection table

The right-hand column of the map makes eight "key facts" and lists twelve cross-connections. Here is each one checked against sources as of October 2026.

Key Facts

Map claim	Verdict	Notes
Nvidia supplies GPUs to virtually all major clouds and AI labs.	**True**	Still true in 2026. Even labs with their own chips (Google, OpenAI, Anthropic, Meta) buy Nvidia. Rubin is live at Azure, AWS, Google, OCI, CoreWeave and Nebius.
Google hosts Grok models via Google Cloud.	**True**	Grok is a managed partner model on Vertex AI / Model Garden [OFFICIAL].
Google leases about 110,000 Nvidia GPUs from SpaceX/xAI.	**True**	Per SpaceX's SEC filing: $920M/month, Oct 2026 to June 2029 [REPORTED]. Counterparty is SpaceX after the xAI merger.
Anthropic uses AWS Trainium, Google TPUs, Nvidia GPUs and xAI Colossus.	**True, incomplete**	Add AMD MI450 (up to 2 GW from 2027) [OFFICIAL].
Grok models are available on Oracle Cloud.	**True**	Since June 17, 2025 [OFFICIAL].
OpenAI uses Azure, Oracle (Stargate) and CoreWeave.	**True, incomplete**	Also AWS, Google Cloud, Cerebras, AMD and its own Jalapeño.
OpenAI and Broadcom are co-developing Jalapeño.	**True**	Revealed June 24, 2026 [OFFICIAL].
Meta uses Nvidia GPUs and custom MTIA chips.	**True, incomplete**	Add AMD (6 GW) and Broadcom co-development through 2029 [OFFICIAL].

Summary of major cross-connections

From → To	Map says	Status, Oct 2026
Google Cloud → xAI (Colossus)	Leases compute capacity	Confirmed; counterparty is SpaceX.
xAI → Anthropic	Compute customer	Confirmed: all of Colossus 1, May 2026.
Google Cloud → xAI (Grok)	Hosts Grok models	Confirmed.
xAI → Oracle Cloud	Grok available on OCI	Confirmed.
OpenAI → Microsoft Azure	Primary cloud partner	**Outdated.** Azure is one of several; OpenAI shifted to renting from Azure, Oracle, AWS and Google [REPORTED].
OpenAI → Oracle Cloud	Stargate partner	Confirmed; Abilene is the only large live site.
OpenAI → CoreWeave	Compute partnership	Confirmed; up to about $22.4B in contracts.
OpenAI → Broadcom	Jalapeño co-development	Confirmed; 10 GW program.
Google → Broadcom	TPU co-development	Long-standing; Broadcom as TPU 8t partner [UNVERIFIED]; MediaTek on 8i.
Anthropic → AWS	Trainium compute	Confirmed and expanded to 5 GW.
Anthropic → Google Cloud	TPU & GPU compute	TPUs confirmed (up to 1M).

From → To	Map says	Status, Oct 2026
Meta → Meta (Infra)	MTIA chips + Nvidia GPUs	Confirmed; plus AMD.

GAPS

10. What the map misses

- **Microsoft Maia 200.** A deployed, 3nm hyperscaler inference chip, absent from the chip layer.
- **AMD's gigawatt deals.** 6 GW with OpenAI, 6 GW with Meta and 2 GW with Anthropic make AMD a major lab supplier, but the map draws no lab-to-AMD lines.
- **Nvidia + Groq.** Groq appears as an independent rival; its technology and leaders now feed Nvidia's roadmap.
- **OpenAI → Cerebras.** A 750 MW+ deal with warrants and a $1B loan, not drawn.
- **Investment loops.** The map deliberately omits investments, but they now shape the whole picture: Nvidia owns part of OpenAI, OpenAI holds warrants in AMD and Cerebras, Amazon and Google own parts of Anthropic, SK Hynix owns part of Etched.
- **China's parallel stack.** DeepSeek V4 on Huawei Ascend, Atlas SuperPoDs, and US export controls that keep Nvidia's H20 and H200 largely out of China [REPORTED]. The map shows Huawei and Cambricon only as a line of text.
- **Power as a layer.** Power and cooling sit in one small box, but in 2026 grid access is arguably the tightest constraint on the whole stack.
- **Stale model names.** See section 2 for the current lineups.

SYNTHESIS

11. Seven forces shaping the stack in 2026

1. Inference is the battleground. Jalapeño, Maia 200, TPU 8i, Etched FIC, Cerebras, SambaNova SN50, Qualcomm AI250 and Nvidia's Groq deal are all inference-first. Training is a one-time cost; serving is a recurring one that scales with users and agents.

2. Gigawatts are the unit of account. Deals are priced in power capacity because power is scarcer than money. Performance per watt has become the headline metric for every new chip.

3. Labs are becoming vertically integrated. OpenAI now has models, data centers (Stargate) and a chip (Jalapeño). Google already had all three. Anthropic and Meta are buying capacity at a scale that gives them co-design influence.

4. Everyone multi-sources. No major lab depends on a single chip or cloud. Anthropic runs on four chip families; OpenAI on at least five. This weakens any one supplier's pricing power.

5. Compute is a tradable commodity. SpaceX renting Colossus to Anthropic and Google, and Google calling its lease "bridge capacity", show compute behaving like a market where even rivals trade.

6. Circular financing. Suppliers invest in customers who buy their chips (Nvidia in OpenAI, AMD warrants to OpenAI, OpenAI loans to Cerebras). It accelerates buildout but ties the companies' fortunes together, which is a risk if demand slows.

7. AI is designing AI hardware. Jalapeño went from design to tape-out in about nine months with model-written kernels. If that holds, custom chips become cheaper and faster to make, which favors more players over one dominant one.

CHRONOLOGY

12. Timeline, May 2025 to October 2026

Date	Event	Layer
June 17, 2025	Grok models launch on Oracle Cloud	Models / Cloud
Sept 2025	Nvidia-OpenAI letter of intent: 10 GW, up to $100B	Chips
Oct 6, 2025	AMD-OpenAI 6 GW deal with share warrants	Chips
Oct 13, 2025	OpenAI-Broadcom 10 GW custom accelerator program	Chips
Nov 6, 2025	Google TPU v7 Ironwood GA	Chips
Dec 2, 2025	AWS Trainium3 GA	Chips
Dec 9, 2025	Agentic AI Foundation formed; MCP donated	Users / agents
Dec 24, 2025	Nvidia licenses Groq technology, hires its leaders	Chips
Jan 2026	OpenAI-Cerebras 750 MW deal; Microsoft Maia 200 (Jan 26)	Chips
Feb 2026	SpaceX completes xAI merger; Nvidia stake in OpenAI reset to about $30B	Cloud / Investment
Mar 16, 2026	GTC: Vera Rubin, Feynman roadmap; Stargate restructured	Chips / Infra
Apr 2026	Muse Spark (Apr 8); TPU 8t/8i (Apr 22); GPT-5.5 (Apr 23); DeepSeek V4 (Apr 24); Anthropic-AWS 5 GW	Models / Chips
May 2026	Anthropic rents Colossus 1 (May 6); Cerebras IPO (May 14); Rubin full production	Cloud / Chips
June 2026	Google-SpaceX GPU lease; Jalapeño revealed (June 24); Etched FIC (June 30)	Cloud / Chips
July 2026	SambaNova at $11B; AMD-Anthropic 2 GW (July 22); Etched $10.3B (July 23); Opus 5 (July 24)	Chips / Models
Aug 2026	Qwen 3.8-Max (Aug 3); Etched about $21B, first rack to Jane Street (Aug 18); Jalapeño at Hot Chips (Aug 25)	Models / Chips
Sept 2026	GPT-6 Astra (Sept 3); GPT-6 Sol/Luna and Claude Opus 5.5 (Sept 22)	Models
Oct 2026	GPT-6 Instant (Oct 7); Gemini 4 Argon GA (Oct 13); Tenstorrent $1.5B (Oct 19); Google lease begins	Models / Chips

REFERENCE

13. Glossary

Term	Meaning
ASIC	Application-specific integrated circuit: a chip built for one job (here, AI math) rather than general use like a GPU.
Inference vs training	Training builds a model from data once. Inference runs the trained model to answer requests, every time a user asks.
HBM / HBM4	High-bandwidth memory: DRAM stacked vertically next to the processor. HBM4 is the 2026 generation.
FP4, FP8, MXFP4	Low-precision number formats (4 or 8 bits). Fewer bits means more operations per second and less memory, at some accuracy cost. MX formats share a scale across small blocks of numbers.
PFLOPS / EFLOPS	Peta (10^{15}) or exa (10^{18}) floating-point operations per second.
TDP	Thermal design power: the power a chip is rated to draw and dissipate.
NVL72, rack-scale	72 GPUs in one rack acting as a single large accelerator over NVLink. “Rack-scale” systems sell the whole rack as the product.
Scale-up vs scale-out	Scale-up links chips tightly inside a rack or pod; scale-out connects many racks over a data-center network.
Mixture of experts (MoE)	A model split into many sub-networks where only a few are active per token (e.g. Qwen 3.8-Max: 2.4T total, 95B active).
Tape-out	The point where a chip design is finished and sent to the foundry to be manufactured.
CoWoS	TSMC’s packaging that places compute dies and HBM side by side on a silicon interposer.
Neocloud	A cloud company focused mainly on renting GPUs for AI, such as CoreWeave, Lambda or Nebius.
Gigawatt (GW)	A billion watts. The 2026 unit for sizing AI data-center deals; roughly a large power plant’s output.
MCP	Model Context Protocol: an open standard for connecting AI agents to tools and data.

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